

What the July 28 Review Changed

Trigger-Agnostic Behavioral Trust for Backdoor-Resilient Federated GPS Spoofing Detection

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The short version: the review asked for five things we had not done. Doing them changed the paper more than any previous week. Two claims we had published turned out to be wrong, and the experiment the review called "the single most valuable" found a real failure in our defense that we had never tested for. The defense is narrower than we thought, and the evidence behind it is much stronger.

1. What we were asked to do

The review listed seven comments. Five required new experiments; two required rewriting claims. We addressed all seven. The experiments are all driven by one shared harness so they cannot disagree with each other, and a single command reproduces every table.

#	What was asked	What it cost us
1, 5	Compare against published defenses, not only our own ablations	Found a baseline that ties us
2	Test uneven (non-IID) client data	Found our defense fails there
3	Stop claiming both defense layers are necessary	Claim withdrawn
4	Separate "removed the attack" from "detector is reliable"	A wrong explanation corrected
6	Recompute the overhead on a realistic denominator	Number was 3x too low
7	Soften four specific overclaims	Applied verbatim

2. The two things we had published that were wrong

2.1 We blamed the dataset for our weak detector. It was our own setup.

We had written that the detector catching only 53% of spoofed signals was "largely a ceiling of this public dataset". The review pushed on it, so we tested it: we trained several models centrally on the same features, with no federation and no attack, which upper-bounds what the federated system could reach.

Model	Spoofing recall	F1	Attack success
Logistic regression	0.2241	0.3146	0.9394
MLP 64-32-16 (paper model)	0.9073	0.8509	0.3779
Random forest (400 trees)	0.9595	0.9392	0.1037
Hist gradient boosting	0.9929	0.9538	0.1213
Federated honest FedAvg (paper baseline)	0.5292	0.5944	0.6367

Every row except the last is trained centrally on the full data pool. The last row is the federated detector used throughout the paper.

The same architecture we use reaches 0.907 recall when trained centrally, and gradient boosting reaches 0.993, against 0.529 federated. The features are separable. Our federated configuration, twelve rounds of three local epochs on a small network, simply underfits. This was our explanation being wrong, not a small correction, and the paper now says so.

2.2 Our overhead figure used the wrong denominator.

We reported 1.1% server-side overhead. That divided the added server time by the sum of every drone's training time, which is what a simulation does. Real drones train at the same time, so a round takes about as long as the slowest one. On that denominator the same measurement is 3.40%.

Defense	Server ms/round	% of sequential	% of parallel
Behavioral trust + median (ours)	34.0	0.98%	3.40%
FLTrust	168.3	4.88%	16.86%

We also had to withdraw the claim that the defense stays practical as the fleet grows. Server cost rises with the number of drones while a parallel round does not, so the overhead fraction climbs from 5.6% at five drones to 34.9% at forty. We only ever tested ten.

3. The biggest finding: the defense does not survive uneven data

The review called this "probably the single most valuable additional experiment", and it was right. Every result we had ever produced assumed each drone sees a similar mix of signals. A real fleet does not. When we tested drones holding different proportions of spoofed data, our trust mechanism stopped working.

Measure	Evenly split	Mild skew	Moderate skew
Trust given to compromised drones (fair share 0.100)	0.0001	0.0942	0.1023
Compromised drones caught	100.0%	8.3%	0.0%
Honest drones wrongly flagged	0.3%	2.1%	0.0%
Backdoor lift, trust layer alone	+0.0039	+0.2374	+0.2482
Backdoor lift, full defense	-0.0265	+0.0647	+0.1002
Backdoor lift, no defense at all	+0.2415	+0.2583	+0.2455

Mean over three seeds. Lift of zero means the attacker gained nothing; the trust layer alone under skew is statistically indistinguishable from having no defense. Read the false-flag row with care: the 0.0% under moderate skew is not an improvement. The gate has stopped firing at all, so it flags nobody, attacker or honest drone alike.

Why it happens

The coordinator scores each drone by how far its answer falls below what the fleet typically answers, measured in units of how much the fleet normally disagrees. When every drone sees similar data the fleet agrees closely, so a backdoored drone's odd answer stands out by a wide margin. When drones legitimately hold different data they disagree more, that margin widens, and the attacker's odd answer no longer stands out. It hides inside the fleet's own variation.

We tested whether this is just a badly set threshold by lowering it. Detection partly recovers, from 8% to 39% under mild skew, which confirms the mechanism, but it never returns to the 100% we get on even data and it starts flagging honest drones instead. So the problem is the way suspicion is scaled, not the threshold value.

This is the result we least wanted and the one we are most glad we ran. Our headline claim is now scoped to fleets with roughly even data, and the clearest next step is a suspicion measure computed per drone rather than against the whole fleet.

4. What the comparison against other methods showed

We implemented every defense the review named and ran them on the identical data, attack, seeds and metrics. Nothing here is quoted from another paper.

Method	Spoofing caught	Backdoor lift	Server ms
Honest FedAvg (no attack)	0.5292	+0.0000	1.0
FedAvg	0.3641	+0.2415	1.0
Accuracy-weighted FedAvg	0.3524	+0.3036	1.3
Coordinate-wise median	0.5008	+0.0646	0.9
Trimmed mean	0.4808	+0.0787	1.4
Krum	0.5168	+0.0331	1.4
Multi-Krum	0.5282	+0.0061	1.4
FLTrust	0.4687	+0.0787	204.4
Behavioral trust (ours)	0.5307	+0.0039	38.0
Trust + median (ours, D2)	0.5560	-0.0265	38.6

Highlighted: Multi-Krum, an existing method, and our full defense. Lower lift is better.

Multi-Krum, a standard existing method, reaches +0.0061 against our +0.0039. At three seeds those are indistinguishable, and it is 27 times cheaper on server time. We report this in the paper rather than omit it, and we removed our earlier claim that a behavioral probe is necessary to stop this attack.

What still distinguishes our method

- Multi-Krum, trimmed mean and Krum must be told how many drones are compromised. A real coordinator does not know that. When we set them to expect two and actually compromise four, Multi-Krum degrades to +0.2837 and trimmed mean to +0.2992, close to no defense at all, while ours holds at +0.0114. Ours has no such setting.
- Those methods produce a selection, not a per-drone judgment, so they cannot report which aircraft is compromised. Ours names them, at 100% detection and a 0.3% false-alarm rate on even data.

5. The claim we withdrew, and what replaced it

We had claimed both defense layers were necessary. The review pointed out that our own numbers do not support it: the trust layer alone reaches +0.0039 and the full defense -0.0265, and at three seeds that gap is inside the run-to-run spread. We withdrew the claim.

The review then asked for the right experiment: degrade the trust score until it is imperfect, and see whether the median layer limits the damage. We ran seven such conditions.

Condition	Trust only	Full defense	Median helped by
baseline (D2, IID)	+0.0039	-0.0265	+0.0305
root 6,000 -> 600	+0.0041	-0.0276	+0.0317
30% root labels flipped	+0.0032	-0.0137	+0.0169
scaling 3 -> 10	+0.0064	-0.0266	+0.0330
dead-zone tau 2 -> 6	+0.0089	-0.0121	+0.0210
Uneven data, mild skew	+0.2374	+0.0647	+0.1727
Uneven data, moderate skew	+0.2482	+0.1002	+0.1480

The last two rows are the uneven-data conditions from section 3, where the trust layer fails outright.

Under mild degradation the median layer contributes between +0.017 and +0.033, which is the same size as our seed spread, so we do not claim it as proven there. Where it clearly earns its place is under uneven data: it removes +0.173 and +0.148 of the attack, roughly ten times more than on even data. So the honest statement is the one the review suggested: the trust layer does most of the work, and the median is a backstop for when it fails.

One condition did not do what we intended, and we report it as such. Making the attack stronger by scaling its update up to ten times instead of three actually weakens it: the undefended lift falls to -0.5888, because an update scaled that hard destroys the shared model rather than steering it. The attack is already near its most effective setting.

6. The numbers that moved

All results are now produced by one harness rather than several notebooks, so the headline table shifted slightly. The conclusions are unchanged; only the last decimals moved.

Quantity	Before	After
Attack, backdoor lift	+0.2457	+0.2415
Attack plus lying, backdoor lift	+0.3036	+0.3036
Full defense, backdoor lift	-0.0253	-0.0265
Full defense, spoofing caught	0.5546	0.5560
Server overhead per round	37.4 ms	34.0 ms
Overhead as a share of a round	1.1%	3.40%

7. Where this leaves the project

Stronger than before

- Nine aggregation rules compared on one pipeline, including every baseline the review named.
- Every number traceable to an exported CSV, and one command reproduces all of them.
- The full operating point reported, not just accuracy: precision, recall, F1, balanced accuracy, false-alarm rate and confusion counts.
- We found our own most important weakness before a reviewer did.

Honestly weaker than we thought

- The headline result is scoped to fleets with roughly even data. On uneven data, which is the realistic case, the trust layer stops firing.
- An existing method, Multi-Krum, matches us on even data at a fraction of the cost. Our advantage rests on not needing to know the attacker count, and on naming the culprits.
- The detector itself is weak, and that is our federated configuration rather than the data.

The three things worth doing next

- Compute suspicion per drone instead of against the whole fleet. This is the direct fix for the uneven-data failure and we already have the harness to test it.
- Train the federated detector properly. Thirty rounds with a larger network reaches 0.851 recall against our 0.529, most of the way to the centralised ceiling.
- Test on a second, independent signal domain. Everything here is one dataset.